Shakil Ahmed Rafi

Ph.D. Candidate,
Aspiring Data Scientist & ML Engineer.
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Passionate and hard-working graduate student working at the intersection of machine learning, deep learning, and differential equations. I aim to bring the tools of mathematics to bring value and make a difference in my community.

Education

University of Arkansas, Ph.D.
2020-Present
Pure Mathematics.
Recipient, Lawrence Jesser Toll Jr. Endowed Scholarship.
Treasurer, Graduate Student Colloquium.

University of Arkansas, M.Sc.
2016-2019
Pure Mathematics.
Vice President, Bangladesh Student Organization

Troy University, B.Sc.
2011-2015
Pure Mathematics & Philosophy.
Graduated Cum Laude.
Member, Alabama Eta chapter of Pi Mu Epsilon, the national math honor society.
Recipient of the Millennium Scholar's Award.

Employment History

PhD Candidate, University of Arkansas
2020-Present
Research area involves the use of artificial neural network in solving certain elliptic PDEs.

Graduate Teaching Assistant, University of Arkansas
2016-2020
Responsible for teaching, lesson-planning, and interfacing with a cohort of faculty in teaching 500+ students along with grading and office hours.

Asst. Business Analyst, R.S.S. Wears Ltd.
2015-2016
Implemented software for a garment manufacturing start-up, including, payroll, book-keeping with Excel, website setup, paper-trail, and staff directories.

Concessions Manager, Sodexo, Inc.
2011-2013
Set up concessions for Troy University games, with a team of six other people, including logistics, safety and customer satisfaction.

Technical Skills

Experienced in Python, including Scikit-learn, XGBoost Pandas, Folium, Geopandas, matplotlib, plotly, NumPy, Pandas, Dask, and other tools.

Proficient in SQL, Microsoft Azure, and PowerBI.

Certified Wolfram Technology Associate. Skilled in MATLAB.

Diligent, collaborative, excellent communication, and oral skills.

Project, Workshops, and Talks

SIAM Mathematics of Data Science, 2022
San Diego, CA
Who Rides Uber Anyway? Rafi & Nithila. We present a clustering approach to classifying ride shares in Chicago during 2020 and the insights it gave us. Pre-print (clickable hyperlink on PDF version): http://doi.org/10.36277/techrxiv.21076042.v2.

Arkansas Summer Research Institute, 2022
Online
Who and where did loans go during the pandemic? Rafi. We looked at loans disbursed during the pandemic (PPP) and saw that poorer neighborhoods tended to get smaller and less frequent loans.
Funded by Arkansas NSF EPSCOR and DART.

Arkansas Spring Lecture Series, 2022
Fayetteville, AR.
Analysis of ride-sharing data for Chicago during the era of the pandemic. Rafi & Nithila. Poster session showing initial work on ridesharing in Chicago.
Funded by NSF and the University of Arkansas.

MAA Regional MathFest, 2015
Troy, AL
Turing Machines and Theorems. Expository talk at Troy University.

MAA MathFest, 2014
Portland, OR
Congruency Algorithm between two regular n-gons. Presented original research at the MAA National MathFest.

References available upon request